Data Visualization & Analytics Lab

Experiment No. : 09

Implement real-time data streaming using Python

Experiment No.09

1. **Aim:** Implement real-time data streaming using Python (with libraries like Plotly or Matplotlib) and visualize the streaming data live.

2. **Objectives:**
   - Understand why visualization is an important part of data analysis.
   - Understand the components involved in visualization design.

3. **Course Outcomes**: Students will be able to understand the challenges and methods for visualizing real-time data.

4. **Hardware / Software Required:** Python

5. **Theory:**
   Streaming data refers to real-time data which is continuously flowing from a source to a target. It includes audio, video, text, or numerical data that is generated by sources such as social media platforms, sensors, and servers.

   Real-time data visualization is the process of displaying data in a visual format as it is generated and processed, allowing users to monitor and analyse data instantly and make quick decisions.

   Real-time streaming data visualization in Python involves displaying data as it's generated, allowing for immediate monitoring and analysis. This can be achieved using libraries like Matplotlib, Plotly, Bokeh, and frameworks like Dash or Streamlit.

   Python Libraries for Data Visualizaion:

   Matplotlib is a fundamental component of Python's Data Visualization toolkit, offering an open-source framework for visually displaying complex patterns in a meaningful manner.

   Seaborn, a Python library that extends Matplotlib, enhances the accuracy of data visualisation. While Matplotlib provides a solid foundation, Seaborn excels at enhancing the visual appeal of statistical graphs.

Geoplotlib enables the creation of various map projections. It's a library for developing interactive maps and geo visualisations.

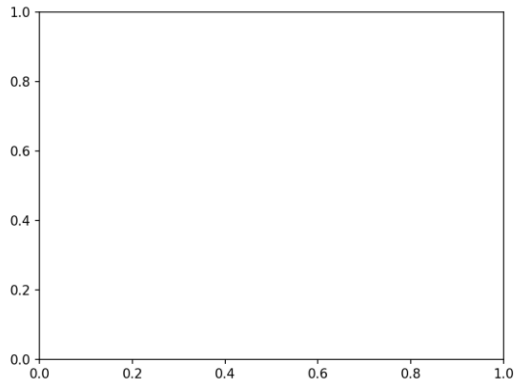
Plotly, a renowned Python Data Visualization Library, is highly regarded for its versatility, enhancing Python's charting capabilities for online settings. The collection encompasses a diverse array of chart kinds, ranging from basic line charts to intricate 3D representations.

Plot real time data using Python:

Let's plot the live CPU usage percentage of PC using matplotlib.

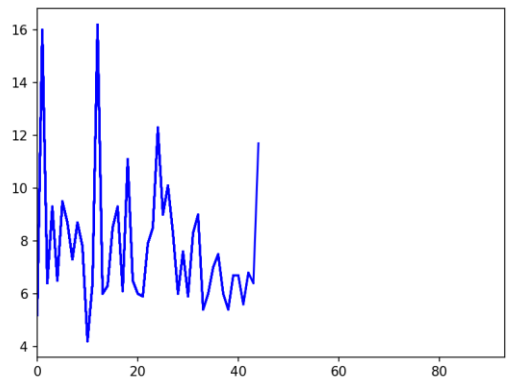
Real_time.ipynb

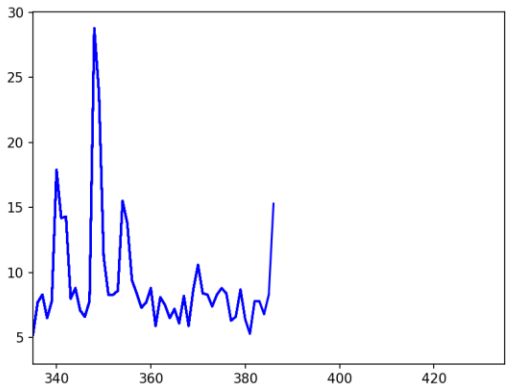
```
import time
import psutil # To get information on CPU usage
import matplotlib.pyplot as plt

#Configure the plot in order to run real time data
%matplotlib notebook  #for creating interactive plots within the notebook
plt.rcParams['animation.html'] = 'jshtml'

fig = plt.figure()
ax = fig.add_subplot(111)
fig.show()
```




```
#infinite loop
i = 0
x, y = [], []
```

```
while True:
    x.append(i)                          #Plot numbers in seq from 0 on x axis
    y.append(psutil.cpu_percent())  #Plot CPU usage percentage on y axis

    ax.plot(x, y, color='b')

    fig.canvas.draw()

    ax.set_xlim(left=max(0, i-50), right=i+50) #Keep moving x axis

    time.sleep(0.1)
    i += 1
```






```
plt.close()
```

8. Conclusion:

We have learnt the use of matplotlib and psutil libraries of python to monitor the CPU usage in real time.

9. Viva Questions:

- What is real time data streaming?
- Which python libraries are used to create the real time data visualization?

References:

1. Visualization with Matplotlib: https://www.oreilly.com/library/view/python-data-science/9781491912126/ch04.html
2. 9 Best Python Libraries For Data Visualization: https://www.index.dev/blog/python-libraries-for-data-visualization